



Title

Basic Hybrid Electric Vehicle (HEV) Power Flow Design

PRESENTED BY

Poonam

NIMS Institute of Engineering & Technology

B.Tech Electrical Engineering

Techn Global Internship Project

June 2026

Hybrid Electric Vehicle (HEV) Power Flow Design Using MATLAB/Simulink

This project explores the design and simulation of a Hybrid Electric Vehicle (HEV) power flow system using MATLAB/Simulink. The focus is on optimizing the interaction between the internal combustion engine and electric motor to enhance efficiency and performance. By analyzing various configurations, we aim to demonstrate the potential of HEVs in reducing fuel consumption and emissions. This work is part of my Techn Global Internship Project at NIMS Institute of Engineering & Technology, scheduled for completion in June 2026.



Define Hybrid Vehicle Configuration

Selected Configuration: Parallel HEV

The parallel HEV configuration allows both the internal combustion engine and electric motor to work simultaneously, enhancing overall vehicle efficiency and performance.

Engine and Motor Synergy

In this setup, the engine and motor collaborate to optimize power delivery, resulting in improved acceleration and responsiveness under various driving conditions.

Key Benefits of Parallel HEV

This configuration significantly reduces emissions and fuel consumption, making it an environmentally friendly choice while maintaining high performance and reliability.



Identify Power Sources



Internal Combustion Engine (ICE)

The ICE serves as the primary power source, providing 60 kW of output, crucial for high-speed performance and long-range driving capabilities.



Electric Motor

The electric motor contributes 80 kW, enhancing acceleration and enabling regenerative braking, which improves overall vehicle efficiency and reduces emissions.



Battery Pack

With a capacity of 20 kWh, the battery pack stores energy for the electric motor, ensuring a reliable power supply and supporting hybrid operation.

Collect Component Specifications

Engine Power Specification

The internal combustion engine delivers 60 kW, ensuring optimal performance and responsiveness during various driving conditions, enhancing overall vehicle dynamics.

Motor Power Specification

With a power output of 80 kW, the electric motor significantly enhances acceleration, providing instant torque and improving the vehicle's responsiveness and driving experience.

Battery Capacity Specification

The battery pack has a capacity of 20 kWh, allowing for extended driving range and supporting efficient energy management in hybrid operations.





System Block Diagram of HEV

This diagram illustrates the power flow architecture of the Hybrid Electric Vehicle. It shows how the engine and motor combine their outputs through a power combiner, optimizing efficiency and performance.

Calculate Energy Flow

1

Engine Output Contribution

The internal combustion engine (ICE) provides a steady output of 60 kW, crucial for maintaining vehicle performance during various driving conditions.

2

Motor Output Addition

The electric motor contributes an additional 80 kW, enhancing acceleration and overall efficiency, especially during low-speed operations and start-stop scenarios.

3

Total Power Output

Combining both sources, the total output reaches 140 kW, ensuring optimal energy flow for effective vehicle operation and improved fuel economy.

Model Torque and Speed Distribution

Power Sharing Dynamics

The model effectively simulates power sharing between the engine and motor, ensuring optimal torque distribution for enhanced vehicle performance under varying conditions.

Balanced Operation

A balanced vehicle operation is achieved through precise torque management, leading to improved acceleration and stability during different driving scenarios.

Performance Metrics

Performance metrics indicate significant improvements in responsiveness and fuel efficiency, showcasing the advantages of hybrid configurations in real-world applications.

Click to add image

Simulate Power Split and Fuel Consumption

1

Simulink Simulation Overview

A comprehensive Simulink simulation was conducted to analyze power distribution between the engine and motor, ensuring optimal performance under various driving conditions.

2

Parallel Power Distribution

The parallel configuration allows simultaneous power delivery from both sources, enhancing overall system efficiency and reducing energy losses during operation.

3

Fuel Consumption Analysis

Efficient energy utilization from the hybrid system leads to a significant reduction in fuel consumption, validating the theoretical models and assumptions made during the design phase.

Analyze Efficiency Under Different Loads

- **Motor Efficiency Considered**

Motor efficiency was maintained at 90% during the analysis, ensuring accurate performance metrics across various load conditions for the hybrid vehicle.

- **Performance Evaluation**

The performance of the HEV was evaluated under different operating conditions, revealing how load variations impact overall efficiency and fuel consumption.

- **Fuel Economy Optimization**

Through optimization techniques, better fuel economy was achieved, demonstrating the importance of efficient power distribution in hybrid electric vehicles.

- **Load Impact Insights**

Insights gained from the analysis highlight how varying loads affect the overall efficiency, guiding future improvements in hybrid vehicle design.

Conclusion

The HEV power flow model effectively illustrates the synergy between the internal combustion engine and electric motor, achieving a total output of 140 kW. This simulation not only enhances understanding of hybrid vehicle dynamics but also highlights the potential for improved fuel efficiency. Future work should focus on real-world applications and further optimization of energy management strategies to maximize performance and sustainability in hybrid vehicles.